Anupreet Singh

anupreet2226579@gmail.com | Portfolio | LinkedIn | GitHub | +1 (410) 260-9818

EDUCATION

University of Maryland, Baltimore County — Baltimore, MD
(Master of Science in Computer Science, GPA: 3.8/4.0) — *Aug 2024 – May 2026*

Indian Institute of Information Technology, Sonepat — Haryana, India
(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10) — *Aug 2019 – May 2023*

SKILLS / RELEVANT COURSEWORK

Programming Languages: C++, C, Swift, Python

Tools and Frameworks: POSIX APIs, launchd, systemd, kqueue/epoll, GCD, SwiftUI, GoogleTest, ThreadSanitizer

Areas of Interest and Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithms

PROJECTS

On-Device Experimentation Service | *A/B assignment daemon and client library in C++ and Swift*

- Built a **C++ experiment-assignment daemon** that **launchd** (macOS) and **systemd** (Linux) start on demand by socket activation, serving client processes over Unix domain sockets from one non-blocking **kqueue/epoll** event loop and bucketing each device by a salted hash of its ID, so it keeps its variant across reboots with **no server round trip**.
- Wrote a C **client library** with a Swift wrapper whose one-call `variant(for:)` reads an `mmap`'d snapshot the daemon swaps atomically via `rename` on each config push, keeping lookups lock-free at **∼90 ns** versus 30 µs over the socket.
- Moved exposure logging off the event loop into a mutex-guarded ring buffer flushed by a writer thread on a coalesced 60 s timer, cutting wakeups **12×** at under **2 MB RSS**, tested under **GoogleTest** and ThreadSanitizer.

Role Radar | *Native macOS and Windows job-alert app with a background checker daemon*

- Designed a native **SwiftUI** macOS app that supervises the checker **daemon** as a **launchd** agent, restarting it within a minute if it stops and giving it 60 s after SIGTERM to drain in-flight work, and reads its state through JSON CLI calls over pipes on **GCD** queues, so non-technical users get alerts with **no terminal or config files**.
- Enforced **one checker per machine** with an `flock`-held PID file (a locked byte on Windows) that the OS frees on any exit, capped concurrent requests with global and per-source **semaphores**, and wrote **380+** unit and integration tests.
- Designed a **laptop-to-Lambda failover** that keeps checks running while the Mac sleeps, electing one runner through a conditional-write DynamoDB lease whose epoch is a **fencing token** on every save, expiring it by wall clock since macOS's monotonic clock stops during sleep, with Lambda taking over within **8 min**.

Rendering Pipelines from Scratch | *Ray tracer, rasterizer, and real-time OpenGL renderer in C++*

- Built a **C++ ray tracer** with recursive reflection and refraction, accelerated by a median-split BVH and a thread pool that claims row chunks off an atomic counter, cutting a 512×512 render from **106 s to 0.14 s (∼759×)**.

INDUSTRY EXPERIENCE

Data AI Engineer | *Platinum Business Services (R&D Team)* | *Catonsville, MD* — Dec 2025 – Present

- Traced **out-of-memory** model-load failures across app, container, and OS layers to 4–5 GB model files lingering in the page cache, and fixed them with `posix_fadvise` eviction on a worker thread, **eliminating the failures**.
- Partnered with compliance analysts to hand-label a **150-control evaluation set** for the RAG compliance agent, benchmarked 5 embedding models (OpenAI, Anthropic, Ollama) on it, and built a priority-ordered fallback that re-runs low-citation queries on the next model's index, raising assessment **accuracy from 64% to 92%**.

Software Development Engineer | *Turfco Landscape Co.* | *Surrey, BC* — Jan 2023 – June 2024

- Architected a **self-serve quoting platform** that priced jobs from admin-editable rules behind a write-invalidated **Redis** cache, with a per-quote audit trail of every rule applied, cutting quote turnaround from **∼2 days to <1 min**.
- Streamed every quote's inputs and fired rules to a retained **Kafka** topic that an analytics consumer joined to bookings, then ran **A/B tests** on price-rule changes over a hashed 50/50 quote split, so the owners priced on conversion data.

ACADEMIC EXPERIENCE

Graduate Assistant | *UMBC - Operating Systems & Data Structures* — Jan 2025 – June 2026

- Designed 10+ **C++** assignments on OS fundamentals (threads with mutexes, semaphores, and lock-free CAS; **IPC** over sockets and pipes; page tables and page-fault handling) and graded them for **40+** students.
- Developed **200+ unit tests** for data-structure projects (trees, heaps, hash tables), cutting grading time by over **80%**.
- Mentored 10 student **Scrum** teams through sprint planning and helped **50+** students debug C++ projects.